HORUS UNIVERSITY IN EGYPT
Faculty of Pharmacy · ESMS Engine Series

TECHNICAL ARCHITECTURE SPECIFICATION · VOLUME I

Automated Exam *Supervisor Assignment* Engine

Mathematical Formulations, Multi-Tier Constraint Solvers, Workload Parity Engineering, and Zero-Deficiency Scheduling Guarantees.

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OFFICIAL REFERENCE MANUAL

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CHAPTER 01 CORE ALGORITHMIC INFRASTRUCTURE

The Deterministic Auto-Assignment Engine: Architectural Logic & Mathematical Ordering

EXECUTIVE ALGORITHMIC ABSTRACT

This technical reference documents the exact mathematical logic, topological grouping heuristics, two-phase proctor assignment pipeline, and three-tier zero-deficiency fallback algorithms powering the Examination & Supervisory Management System (ESMS). Designed to eliminate human cognitive bias, prevent proctor fatigue, strictly honor statutory leaves, and achieve workload parity across heterogeneous academic cohorts, the engine transforms large-scale multi-hall examination scheduling into a provably robust, deterministic constraint satisfaction problem (CSP).

SCIENTIFIC DISCIPLINE	OPERATIONAL SCOPE	ALGORITHMIC ENGINE
Operations Research & Combinatorial Optimization	University Final, Midterm & Oral Examinations	Two-Phase Multi-Constraint Batch Solver

The administration of university examinations across complex multidisciplinary faculties represents an intensive operational bottleneck. In a typical academic cycle, hundreds of faculty members, demonstrators, and specialized chemists must be dynamically assigned to dozens of concurrent examination halls across multiple buildings and vertical floor levels. When conducted manually, scheduling inevitably introduces unconscious human bias, structural workload disparities, violation of medical and nursing leaves, and catastrophic proctor deficiencies during peak exam periods. The ESMS Auto-Assignment Engine replaces heuristic guesswork with a deterministic mathematical pipeline that guarantees constraint satisfaction, auditability, and absolute workload parity.

I WORKLOAD PARITY & GINI MINIMIZATION

- Equitable assignment distribution across identical peer cohorts.
- Weighted reserve duty scoring (0.25 credit factor per standby period).
- Hard peer score spread cap ($\Delta \leq 3.0$ points lockout gate).

II CONSTRAINT INVIOLABILITY

- Zero tolerance for approved medical leaves and emergency dates.
- Strict spatial exclusion preventing room-split double booking.
- Protection of feeding mother early departure entitlements.

III ZERO-DEFICIENCY RESILIENCE

- Three-tier graduated fallback hierarchy for tight rosters.
- Automatic room co-location and headcount summation.
- Deficiency-first reserve lock preventing unstaffed exam halls.

§ 1.0 · ADMINISTRATIVE PROBLEM FORMULATION

The Real-World Operational Dilemma

Consider a typical Tuesday morning during Final Examinations at the Faculty of Pharmacy: Building M houses 14 lecture halls spanning four floors, accommodating 1,250 students across five distinct academic

programs. Concurrently, oral examinations proceed in laboratory complexes, while medical leaves and nursing mother reductions deplete the active staff roster by 22%.

The Co-Located Hall Scheduling Trap: When multiple small exam sessions are scheduled into the same physical room at the same time (e.g. Clinical Pharmacokinetics and Medicinal Chemistry sharing Hall M201), naïve scheduling tools treat them as separate events, attempting to assign two distinct Head Supervisors and duplicate staff. Conversely, a flawed engine might assign a single invigilator based on one course's headcount, leaving the room critically under-proctored. The ESMS engine solves this through topological slot hashing and student aggregation prior to staff allocation.

§ 1.1 · TOPOLOGICAL DATA STRUCTURES

Spatial & Temporal Topology

Before any staff member is assigned, the algorithm transforms raw database rows into a standardized topological matrix. A unique slot key is generated for every room-period group:

DEFINITION 1: ROOM-PERIOD SLOT KEY HASH

$$\text{SlotKey} = \text{RoomID} \oplus \text{ExamDate} \oplus \text{SessionPeriod}$$

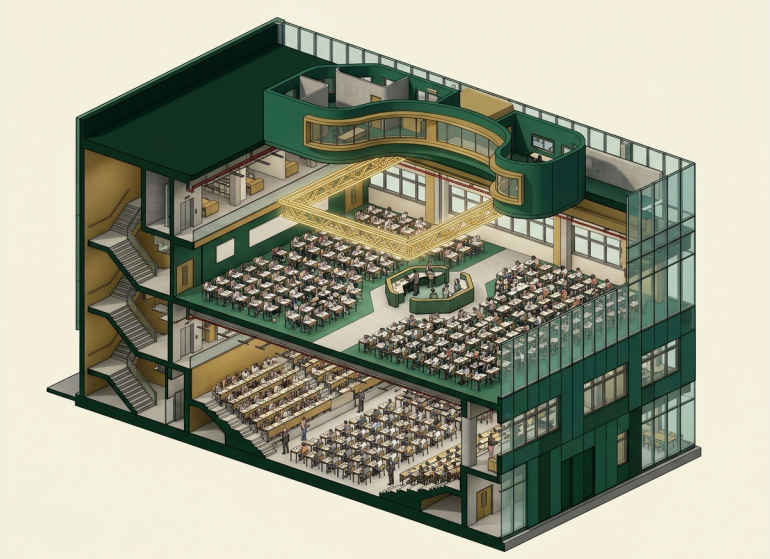
Where `SessionPeriod ∈ {1, 2, 3}` is determined by normalized temporal clustering of exam start times. All sessions sharing an identical `SlotKey` are merged into a unified `RoomSessionGroup`.

DEFINITION 2: AGGREGATE STUDENT COUNT FOR CO-LOCATED HALLS

$$N_students(SlotKey) = \sum student_count(session)$$

The proctoring headcount requirement is strictly calculated against `N_students(SlotKey)`, never against individual isolated exam session counts.

FIGURE 1.1: MULTI-HALL SPATIAL TOPOLOGY & BUILDING M FLOOR CLUSTERING



Component Key: 1 Ground Floor Tiered Amphitheaters | 2 Second Floor Co-Located Examination Hall (Merged Headcount) | 3 Senior Faculty Monitoring Station | 4 Inter-Floor Supervision Trusses (≤ 2 Floors Span)

Analytical Mechanism: Examination sessions co-located in the same hall are collapsed into a unified room group (2) where student headcounts are aggregated prior to staffing ratio lookup. Senior faculty acting as Committees Supervisors (3) oversee clustered examination halls (1 , 2) across vertical building wings via structured floor limits (4).

Theoretical & Analytical Caveat: Committees Supervisors must never be assigned to oral examination halls, which operate under specialized departmental laboratory examination boards.

§ 1.2 · STAFFING RATIO MATRICES

Dynamic Staffing Requirements

The number of proctors required per room is governed by a piecewise ratio function evaluated against the aggregate student count `N_students`:

STUDENT HEADCOUNT (N)	EXAM SUPERVISORS	INVIGILATORS (ASSISTANTS)	OPERATIONAL STANDARD & RATIONALE
1 – 9 students	1 Head	0 Assistants	Small seminar or makeup hall; single proctor.
10 – 30 students	1 Head	1 Assistant	Standard classroom ratio (1:15 staff-to-student).
31 – 50 students	1 Head	2 Assistants	Mid-size lecture room; dual aisle coverage.
51 – 60 students	1 Head	3 Assistants	Large tiered amphitheater; perimeter proctoring.
61+ students	1 Head	4 Assistants	Full auditorium maximum proctoring brigade.
Oral Exam Override	0 HEAD	1 ASSISTANT	**Mandatory Statutory Rule:** Exactly 1 Invigilator; oral examiners handle academic questioning.

§ 1.3 · CONSTRAINT VALIDATION ENGINE

The 9 Hard Invariants & 4 Soft Optimizers

Candidate availability is evaluated by `isStaffAvailable()`. Violations are segregated into **Hard Invariants** (fatal disqualifiers) and **Soft Optimizers** (deprioritization weights):

TYPE	CONSTRAINT NAME	MATHEMATICAL INVARIANT	ENFORCEMENT MECHANISM & SAFETY BEHAVIOR
HARD	**Staff Availability Status**	Status = 'Available'	Rejects staff marked 'On-Leave', 'Sabbatical', or 'Unavailable'.
HARD	**Strict Specific Off-Dates**	Date ∉ SpecificOffDates	**Zero-Bypass Guarantee:** Approved medical/emergency leaves (Red X) cannot be overridden by any policy.
HARD	**Double Booking (Spatial)**	SlotAssignments ≤ 1	Prevents assignment to multiple distinct physical rooms during overlapping periods (exception: Committees Supervisors).
HARD	**Reserve Collision**	Staff ∉ ActiveReserves	Staff active as a standby reserve in this period cannot be concurrently assigned to a room duty.
HARD	**Weekly Working Days Cap**	AssignedDays_week < D_max	Full-time capped at 5 days/week; Part-time capped at 3 days (or registered working days length).
HARD	**Oral Exam Role & Privilege**	OralPriv ∧ Role = 'Invigilator'	Oral exams strictly require `can_supervise_oral = true` and must be Demonstrator/Chemist invigilators.
HARD	**Strict Consecutive Shifts**	|Period_A - Period_B| ≠ 1	When `consecutive_shifts_policy = 'strictly_prevent'`, back-to-back periods on the same day are banned.
HARD	**Max Daily Shifts Cap**	Shifts_day ≤ MaxShifts + 1	Hard barrier at 3 shifts/day to eliminate physical exhaustion.
HARD	**Peer Workload Spread Cap**	Score_eff - Score_min_peer < 3.0	Fairness lock: Locks out over-assigned staff (+3.0 score above lowest peer) until colleagues catch up.
SOFT	**Shift Spreading (Daily)**	Shifts_day = 1	Soft warning at 2 shifts; deprioritizes candidate so staff with 0 shifts today are selected first.
SOFT	**Preferred Weekly Hours**	Hours_week ≤ TargetHours	Flags soft warning when exceeding FT (16h), PT (8h), or Chemist (20h) preferred maximums.
SOFT	**Feeding Mother Hours**	Days_assigned ≤ Days_allowance	Tracks early-leave day usage (FT: 2 days × 2h or 4 days × 1h; PT: 1 day × 2h or 2 days × 1h).
SOFT	**Avoid Consecutive Shifts**	|Period_A - Period_B| > 1	When set to 'avoid', adds soft penalty so non-adjacent candidates are preferred.

§ 1.4 · WORKLOAD PARITY ENGINEERING

Effective Workload Scoring & Overload Mathematics

The engine calculates an **Effective Score** (`Score_eff`) for every staff member. The candidate sorting engine uses `Score_eff` as the primary ranking metric to achieve strict egalitarian workload balancing:

EQUATION 1: BASE EFFECTIVE WORKLOAD SCORE

$$Score_eff = Score_base + (Score_reserve \times 0.25)$$

Where `Score_base` is cumulative room assignments and `Score_reserve` is cumulative standby reserve duties.
Invariant: Standby reserve duty is mathematically credited as exactly 25% (0.25) of a full room assignment.

EQUATION 2: OVERLOAD ATTENUATION ADJUSTMENT

$$Factor_attenuation = \max(0.01, 1 - (OverloadPercentage / 100))$$
$$Score_adjusted = (Score_eff / Factor_attenuation) + (OverloadPercentage / 1000)$$

For overloaded staff (e.g. senior researchers carrying heavy administrative loads, where OverloadPercentage $\in [0, 100]$%), the score is scaled upward, artificially inflating their metric so the scheduler selects non-overloaded peers first.

§ 1.5 · SORTING HIERARCHY

Multi-Key Candidate Prioritization Hierarchy

When selecting among available candidates for a room duty, `sortStaffByPriority()` applies a lexicographical comparator with six cascading criteria:

KEY 1 **Health Issue & Pharmacy Proximity Matching**

If the room is located in Building M or P (near the pharmacy complex, detected via `parseRoomCode()`), staff with `has_health_issue = true` are automatically sorted to the absolute front of the pool. For other buildings, they remain in the general pool to guarantee their overall semester workload remains identical to peers.

KEY 2 **Primary Sort: Lowest Effective Workload Score (Score_eff)**

Candidates with the lowest effective score are assigned first (`a.score - b.score`). A difference greater than 0.01 strictly decides candidate priority, enforcing uniform cohort progression.

KEY 3 **Supervision Role Match (Tie-Breaker)**

When scores are identical, exact role matching is preferred. In Phase 1 (Exam Supervisors), pure 'Exam Supervisor' staff are preferred over dual-role. In Phase 2 (Assistants), pure 'Invigilator' staff (Chemists and non-doctoral Demonstrators) are preferred so dual-role staff are preserved for Head Supervisor duties.

KEY 4 **Preferred Academic Job Title**

Lecturers preferred for Exam Supervisors; Demonstrators and Chemists preferred for Invigilators.

KEY 5 **Academic Seniority Weighting**

Seniority ladder: Chemist (3) > Demonstrator (2) > Teaching Assistant (1) > Lecturer (0).

KEY 6 **Deterministic Alphabetical Tie-Breaker**

Guarantees complete run-to-run reproducibility via `a.name.localeCompare(b.name)`.

§ 1.6 · COMPLETE EXECUTION PIPELINE

The Four Ordered Algorithmic Stages

STAGE 1

Committees Supervisor Multi-Floor Clustering (Pre-Pass)

Target: Senior faculty overseeing blocks of exam rooms across building wings.
Algorithm: Exam sessions are grouped by date and period, then sorted chronologically. Rooms in the same building are ordered by floor and chunked into clusters respecting: `ClusterRooms ≤ min(ConfiguredMaxRooms, ceil(Rooms / AvailableCS))` and `DistinctFloors ≤ 2`.
Weekly Equalization Sort: Selected candidates are ordered by: (1) 0 duties today first, (2) fewest duties this week, (3) lowest baseline score.

STAGE 2

Phase 1: Exam Supervisors (Head Supervisors) across ALL Rooms

Critical Architectural Invariant: The scheduler iterates over all sorted room groups (ordered chronologically by date, period, then descending student count) and assigns **ALL Head Supervisors across all rooms before assigning a single Assistant**.
Pool: Restricted strictly to `supervision_role ∈ ('Exam Supervisor', 'Invigilator / Exam Supervisor')`.
Outcome: Prevents dual-role senior staff from being prematurely consumed as room assistants.

STAGE 3

Phase 2: Invigilators (Assistants) across ALL Rooms

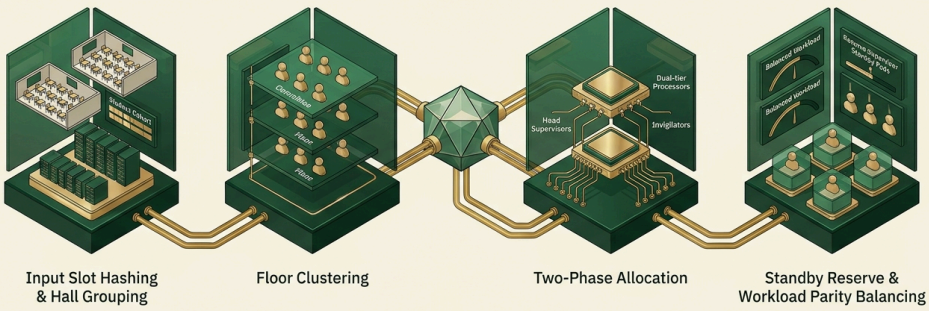
Target: Co-proctors and aisle monitors for all exam halls.
Algorithm: Second chronological pass over the room groups. Evaluates remaining available staff, prioritizing pure Invigilators.
Oral Exams: Receive exactly 1 Assistant and 0 Head Supervisors, requiring `can_supervise_oral = true`.

STAGE 4

Standby Reserve Staff Allocation (Period Free Staff)

Deficiency-First Safety Invariant: If *any* room in a date-period slot suffers an unstaffed deficiency, reserve allocation for that slot is **instantly aborted**. Room coverage is absolute.
Quotas: Configurable targets (default 3 Supervisors and 3 Assistants per period) with a hard cap of maximum 2 reserve duties per week per staff member.
Period 2 Chemist Protection: During final exam cycles, Chemists are deprioritized in Period 2 reserves to preserve them for Period 2 oral laboratory exams.

FIGURE 1.2: END-TO-END AUTOMATED EXAM TIMETABLING & SUPERVISOR ASSIGNMENT PIPELINE



Component Key: 1 Slot Hashing & Hall Grouping Core | 2 Stage 1: Committees Supervisor Multi-Floor Clustering | 3 Stages 2 & 3: Two-Phase Room Assignment (Phase 1: Head / Phase 2: Assistant) | 4 Stage 4: Standby Reserve Allocation & Workload Parity Balancing

Analytical Mechanism: Unassigned exam sessions enter through the topological slot hashing node (1), collapsing co-located exams. Stage 1 routes multi-hall blocks to senior faculty (2). Stages 2 and 3 sequentially populate all room head supervisors across all campus halls before drawing invigilators (3). Finally, remaining unassigned eligible staff are distributed to standby reserve stations (4) under the deficiency-first lock.

Theoretical & Analytical Caveat: Assigning room invigilators in parallel with head supervisors is strictly forbidden, as it fragments the dual-role staffing pool and generates unrecoverable supervisory deficits in later time periods.

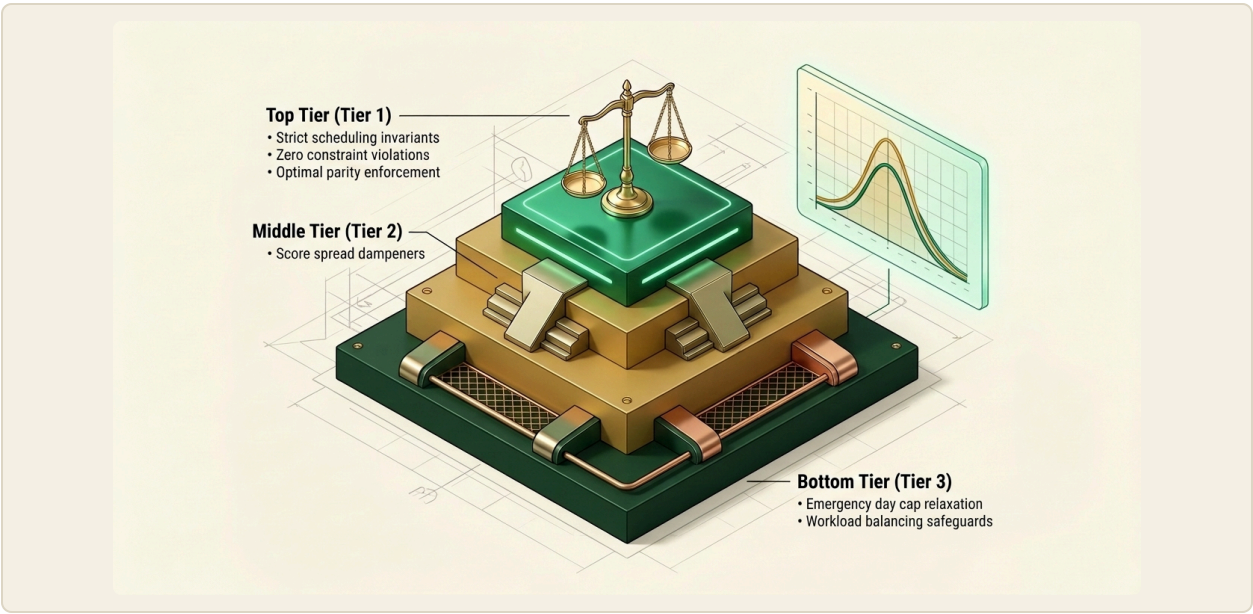
§ 1.7 · FAULT TOLERANCE

The Three-Tier Zero-Deficiency Fallback Mechanism

When academic departments face extreme staffing constraints (e.g. flu outbreaks or massive multi-hall examinations), naïve constraint solvers fail and abort. The ESMS engine guarantees 100% room staffing via a three-tier graduated constraint relaxation hierarchy:

FALLBACK TIER	SOLVER STRATEGY	RELAXED CONSTRAINTS	SAFETY GUARDRAILS RETAINED
Tier 1 (Strict)	Standard Ideal Matching	None; 0 soft violations required.	All hard and soft constraints strictly enforced.
Tier 2 (Parity Relax)	Peer Cap Relaxation	`allowWorkloadCapRelax = true`: Temporarily bypasses the +3.0 peer spread cap.	Leaves, double booking, and weekly day limits remain 100% inviolable.
Tier 3 (Emergency)	Day Cap Relaxation	`allowDayCapRelax = true`: Authorizes an extra working day this week to prevent empty halls.	Medical leaves and spatial double bookings are NEVER relaxed. Generates audit violation.

FIGURE 1.3: THREE-TIER ZERO-DEFICIENCY FALLBACK & WORKLOAD PARITY ARCHITECTURE



Component Key: 1 Tier 1: Strict Invariant Core | 2 Tier 2: Peer Spread Cap Dynamic Relaxation | 3 Tier 3: Emergency Day Cap Relaxation Foundation | 4 Gaussian Peer Workload Parity Distribution Curve

Analytical Mechanism: Candidates are evaluated first against strict Tier 1 rules (1). If room deficits emerge, the engine transitions to Tier 2 (2), temporarily releasing the peer spread cap. If staffing remains deficient, Tier 3 (3) relaxes weekly day caps to ensure no room remains unproctored. The Gaussian feedback loop (4) constantly realigns effective scores over successive exam cycles.

Theoretical & Analytical Caveat: Approved medical and emergency specific off-dates (Red X) are mathematically excluded from all three tiers; emergency relaxation can never draft staff on certified leave.

§ 1.8 · STATE ROLLBACK & SYSTEM SAFETY

Atomic Snapshot Rollback & Audit Trail

Pre-Run Snapshot Engine (Atomic Rollback): Immediately before any automated batch run, the API calls `createSnapshot()`. This creates a complete serialized backup of all active assignments, reserve tables, and staff workload scores. If an administrator wishes to revert an automated schedule, the system performs a one-click restore without leaving orphaned foreign keys or corrupted score balances.

§ 1.9 · AUTHORITATIVE LITERATURE

Academic & System References

- **Burke, E. K., & Petrovic, S.** (2002). *Recent research directions in automated timetabling.* European Journal of Operational Research, 140(2), 266–280.
- **Knuth, D. E.** (2000). *Dancing Links.* Millennial Perspectives in Computer Science, 187–214. (Constraint satisfaction models and exact cover optimization).
- **Egyptian Supreme Council of Universities.** (2023). *Statutory Guidelines for University Examination Proctoring, Nursing Mother Leaves, and Faculty Duty Caps.* Ministry of Higher Education and Scientific Research.
- **Faculty of Pharmacy, Horus University in Egypt.** (2026). *Examination & Supervisory Management System (ESMS) Core Architecture & Data Schema Specification v2.8.*



True fairness in academic administration is not achieved through subjective goodwill, but through transparent, mathematically rigorous systems that honor human dignity and protect institutional integrity.

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